NeuroShard: Native Settlement for Continual, Verifiable Language Model Evolution

Protocol working draft and reproducible experiments

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Abstract

NeuroShard aims to maintain a shared language model through contributed computation, curated data and native token incentives. Its own bonded BFT consensus orders work and payments; execution acceptance and model-quality promotion are separate decisions. We implement experimental full-model pipeline training, identity depth growth, native data admission, challengeable evaluation, serving decisions and paid generation, with explicit assumptions for verification and artifact availability.

A 134.5M-parameter seed trains across three workers on two CPU hosts; identity residual expansion produces 148.7M parameters while preserving tested initial logits. Earlier quality experiments fail promotion gates or remain inconclusive. A compact optimizer extension refutes a tested inconsistent SGD assertion with 19,521 bytes instead of 184,644,776 bytes of full-stage replay input. It increases warm training time by 37.4% and does not detect a coherent invented gradient; complete neural replay remains necessary. A funded integration candidate prepays complete-replay services and preserves transactions and worker state across interruption, while sponsor selection and collusion retain an honest-observer assumption.

Controlled GPU experiments demonstrate narrower useful learning. In a four-worker 1.7B comparison, exact answers on 1,024 generated tasks rise from 105 for the seed to 641 for single-GPU training, 613 for synchronized four-worker training and 557 for local training windows. Local windows use 6.22% of DDP's transmitted bytes and reproduce every worker's uninterrupted model, outer optimizer and Adam/RNG state after a process crash. They fail retention and the declared quality margin against DDP, and remain more expensive than one GPU. Four serving replicas deliver 3.00 times aggregate throughput on a fixed development workload and complete all requests after one provider stops. All models still fail invoice arithmetic. A separate training-only compressed-gradient probe resolves a GPU memory failure but does not establish full-length quality or recovery.

The released public testnet remains a bounded adapter-training and paid-inference demonstrator under concentrated ownership. Independent auditing, adversarial admission, sustained broad quality gains, economical training scale and operational integration remain open. The experiments establish reproducible components and measured tradeoffs; perpetual improvement and a production permissionless compute economy remain unproven.

1 Objective and scope

NeuroShard's objective is an ongoing public learning service. Participants contribute training, inference, verification, or storage; its own blockchain settles duties and payments. The model should be able to consume fresh licensed data, improve on measured tasks, and gain useful capacity when additional resources become available. A particular seed checkpoint is an initial condition, not a permanent size ceiling.

Four properties must be distinguished. Execution correctness asks whether the prescribed computation occurred. Consensus asks which state transition and payment became final. Learning quality asks whether a candidate improves a chosen task distribution. Resource scaling asks whether a larger computation can run at an acceptable cost. A signature proves none of the latter three, and lower training loss does not establish general intelligence.

This draft supersedes the earlier five-page manuscript, preserved with the private historical artifacts. It specifies the target protocol separately from the implemented experiments. The released 0.4.0 public testnet remains a bounded adapter-training and paid-inference demonstrator. The new code resides in `neuroshard.evolution`; its experiments do not silently migrate public balances or replace the public serving model.

2 Related foundations

DiLoCo reduces synchronization between complete model replicas; it does not eliminate communication between the layers of a model-parallel replica or establish Byzantine correctness [2]. SWARM studies training pipelines over heterogeneous, unreliable resources [3]. We use these as distributed-systems foundations, while treating adversarial execution as a separate problem.

Function-preserving model expansion follows the general motivation of Net2Net [4]. Our implemented transformation is specifically an identity residual extension for a tied-weight Llama decoder. It neither reproduces every Net2Net transformation nor establishes that extra depth improves downstream quality.

Verde studies refereed delegation of machine-learning computation under an at-least-one-honest-provider assumption and uses reproducible operators to address hardware-dependent arithmetic [5]. Our complete referee is much coarser: it replays an entire bounded worker stage. The optimizer extension below adds one narrow refutation path. The verifier's dilemma is directly relevant: when correct verification is costly, rational participants can prefer to skip it [6]. Fraud bounties alone are not a demonstrated sustainable audit market. Proof of Diligence explicitly analyzes collusion and shared replay costs [9]; peer-prediction mechanisms study richer reports under assumptions about participants' information and behavior [10]. Neither is a theorem establishing an independent NeuroShard audit market.

3 Roles, assumptions, and state

3.1 Roles

The protocol has validators, task coordinators, compute workers, auditors, data contributors, storage replicas, inference providers, and customers. An account may perform several roles. The number of keys is never treated as evidence of independent operators. Coordinators assemble a job and may be replaced after an objective lease deadline. A public implementation must support competing coordinators and worker discovery without a mandatory operator-owned endpoint.

3.2 Assumptions

Native consensus uses bonded BFT membership and assumes less than one third Byzantine voting weight, with eventual synchrony for progress [7]. Signatures and content hashes are assumed secure. An optimistic execution claim is sound only if an honest, online observer checks the relevant graph, obtains its inputs, and can complete a dispute before its deadline. Sampling fewer stages gives a probabilistic condition, not full coverage by definition.

Learning evaluation additionally assumes sufficiently representative, uncontaminated data. Publicly available tests can be rehearsed off-chain. Committing a model before choosing a public test subset does not prevent a participant from training on the entire public pool. Independent curation, changing tasks, exposure accounting, and external evaluation are therefore part of the quality mechanism.

3.3 State commitments

The target state contains accounts, validator bonds, active task reservations, work commitments, disputes, data manifests, model revisions, evaluation tickets, serving aliases, inference escrows, and emission counters. A model revision commits architecture, component tensor roots, numerical semantics, tokenizer identity, and ancestry. A semantic computation identifier should commit weights, architecture, data, objective, and optimizer settings independently of arbitrary history or job nonces; it prevents rewarding a cached transition repeatedly after a rollback.

The learning head and serving head are separate. Correctly executed training may advance a learning candidate without changing customer-facing inference. Only the quality-promotion transition changes the serving alias. Rejected candidates and unsuccessful work remain inspectable in history. Rollback changes the alias; it does not delete blocks, reverse paid valid work, or authorize paying again for the same semantic computation.

The implemented paid-task identifier commits parameter-component bytes, architecture, effective token inputs and targets, and optimizer settings. It normalizes equivalent explicit/implicit target masks and numeric configuration representations, and omits model ancestry, allocation layout and job nonces. The ledger rejects a task already paid under that identifier. This matters even without rollback: at an exact fixed point, unchanged weights can otherwise be wrapped in endlessly new model-manifest ancestry and rewarded again. A regression test executes this case and permits only its first payment. The identifier is an explicit protocol equivalence rule, not a solution to arbitrary program equivalence or proof of fresh physical energy expenditure.

4 Data as a versioned protocol input

A dataset identity is a content digest, not a mutable S3 key or repository branch name. Source manifests name the repository, full revision, split, license, parser/tokenizer profile, and cursor range. Object replicas may use S3, ordinary HTTP, or peer storage; none is the definition of the dataset.

The implemented collector uses pinned Parquet inputs with checksum verification, a durable cursor, immutable document and sequence objects, normalized document identifiers, and a SimHash near-duplicate screen. Exact document grouping precedes token-window assignment. The SimHash screen is a heuristic; it does not prove absence of paraphrases or semantic overlap. Training and protected evaluation roles share an exclusion registry within a corpus.

Each epoch collects a bounded new window, mixes it with historical replay, and commits the resulting batches. The current replay fraction is 25% when historical data exists. A daily resource budget can renew without erasing the cursor, model ancestry, or ledger. Exhausting a source causes a pause or a separately authorized source addition, not fabrication of fresh examples. A new source must supply provenance and an admissible license. Permissionless submission does not imply automatic acceptance of arbitrary text.

4.1 Assistant-response objectives

A first experiment used 64-token conversation prefixes and scored every next token. In its three evaluation groups, only 17, 18, and 10 of 64 examples contained any first-response tokens. This was an inadequate basis for a claim about improved assistant answers.

The preparation used for the reported paired quality experiments keeps up to 64 preceding prompt tokens and 64 tokens of the first assistant response. It pads only after the response. Labels equal the actual next token for response positions and the explicit ignore marker elsewhere. Every row must contain at least one scored target; invented target tokens and all-ignored rows are rejected. This remains a truncated-context response-loss experiment, not a general dialogue benchmark.

For token sequence x and target mask m, the loss is

$$L(\theta; x, m) = -\frac{\sum_{t=1}^{T-1} m_t \log p_\theta(x_{t+1} \mid x_{\le t})}{\sum_{t=1}^{T-1} m_t}. \tag{1}$$

Both x and m are committed task inputs. A worker cannot choose an easier mask after observing its outputs.

4.2 Content-addressed text semantics

The maintained epoch implementation now binds a model to a serialized fast-tokenizer backend, exact vocabulary and special-token definitions, chat template, bounded rendering rule, and tokenizer-library versions. A SHA-256 codec root appears in the model manifest, corpus settings and prepared windows. Equal vocabulary size does not establish equal token meaning. Training and identity depth growth preserve this root; mixed-codec batches and an implicit vocabulary replacement are rejected. Loading a codec reconstructs and reserializes its backend, and verifies runtime and canonical identity.

The supported template subset excludes clock access, helper calls, filters, nested or recursive loops and object introspection. Message loops and concatenation are bounded, and rendering streams through an output-size limit. Message content cannot contain reserved chat-control tokens. Encoding adds no second set of special tokens and performs no implicit truncation. These rules establish reproducible framing for the supported seed, not compatibility with every template or resistance to semantic prompt injection.

For each nonempty assistant turn, the new preparation requires a stable generation prefix and an actual end-of-response token. It divides real response targets into bounded windows, retaining preceding conversation/answer tokens as context and masking prompt and padding positions. Chunk boundaries do not invent end-of-response targets. Within a document's prepared windows, each retained target appears once. A finite budget rotates through first chunks of all turns before later chunks and records omitted targets. Training may retain such a partial document; evaluation ingestion rejects one whose response targets cannot all fit within the declared budget. This produces a measurable length-selection bias, and complete target coverage still uses truncated context.

Evaluation now reserves whole documents, including all their windows. If window j in document d has n_{dj} scored causal targets and mean loss ℓ_{dj}, its document observation is

$$\ell_d = \frac{\sum_j n_{dj} \ell_{dj}}{\sum_j n_{dj}}.$$

The paired quality gate receives one observation per document rather than treating correlated chunks as independent examples. Parent and candidate must use the same codec for this token-loss comparison. A future vocabulary migration requires explicit embedding/output-head conversion,

retokenization of retained raw data, compatible clients, and evaluation on common raw-text or task outcomes; per-token losses across different tokenizations are not directly comparable. Continuous learning of new concepts does not itself require a growing vocabulary.

These are implemented text and data rules. Native settlement still checks prescribed numerical batches; it does not independently certify their raw-text provenance or usefulness. The quality measurements later in this paper retain their historical first-response objective and are not reinterpreted as results of this newer document-window policy.

5 Full-model training across workers

5.1 Ownership and schedule

The seed is the Apache-2.0 SmolLM2-135M-Instruct checkpoint at revision `12fd25f77366fa6b3b4b768ec3050bf629380bac` [8]. It contains 134,515,008 trainable parameters. The implementation imports its pinned files into float32 tensor components: one embedding, one final norm, and thirty transformer blocks. All parameters participate in training; there is no frozen backbone plus a small adapter in this experiment.

The first worker owns the tied embedding and output head, the final norm, and an initial contiguous block range. Later workers own successive contiguous block ranges. Forward activations travel through the workers; the final hidden state returns to the first worker for the language-model loss. Its output-head gradient travels backward through the stages. The first worker accumulates its output-head and input-embedding contributions into the same parameter tensor.

Keeping the tied head with the embedding avoids sending its dense weight gradient across the network every step. It also creates an asymmetric first stage and a lower bound on that worker's capacity. Placement must account for this asymmetry; summing advertised memory alone is insufficient.

5.2 Optimizer and numerical contract

The current optimizer is clipped SGD, chosen to make state and replay explicit. For worker gradients g_j, the norm and update are

$$n^2 = \mathrm{fsum}_j \, \|g_j\|_2^2, \tag{2}$$

$$a = \min\left(1, \frac{c}{\sqrt{n^2} + 10^{-6}}\right), \qquad \theta'_j = \theta_j - \eta a g_j. \tag{3}$$

Each stage sums squared gradients in float64 with an explicit parameter order; the resulting scalar is committed as a hexadecimal float. The whole optimizer state is the weights and prescribed step policy; Adam moments are not silently omitted from an Adam implementation.

The tested profile uses CPU float32, single-thread execution, disabled MKLDNN, deterministic algorithms, a default ATen CPU path, and MKL SSE4.2 instructions. Package startup establishes this profile before helper modules initialize PyTorch. An early native-chain test exposed the importance of this order: all validators agreed with each other but disagreed with correctly configured workers. The corrected entry point reproduces the reference trace. Agreement alone is not numerical conformance.

The current bounds are 48 million resident parameters per stage, at most 64 stages, at most four rows and 512 total batch tokens, and at most 256 tokens in a row. These are execution-profile bounds, not proofs of a universal memory peak. The participant still needs room for gradients, activations, serialization, and runtime overhead. Other hardware and kernels require an explicit compatible numerical profile and conformance tests.

5.3 Recovery

Workers durably record operation intents and results. An operation identity binds session, step, phase, parent, and inputs. A repeated request returns the committed result; a conflicting request cannot occupy the same operation position. After interruption, a worker reloads the last complete parameter checkpoint and rebuilds only the unfinished graph. Coordinator journals retain the pending batch so a retry cannot silently train a different example. Tests cover a lost acknowledgment after backward computation and resumption through the next step.

Replacement workers need the same committed components and numerical semantics. Data replication is therefore a prerequisite for recovery, not an optional optimization. Current cross-host experiments use authenticated loopback RPC through SSH. They do not constitute a public worker-discovery protocol.

6 Verification and native settlement

6.1 Reservation and compact claim

A coordinator first reserves the current parent and training round on-chain, naming the reward identities. The reservation locks collateral and expires objectively. A submitted claim includes a model transition, a compact execution transcript, the assigned batch, and signed stage receipts that bind the reservation. Relaying a published transcript cannot replace the preassigned reward identities. Replay proves a result, not which physical processor performed it; subcontracting is compatible with the contract.

The ordinary state-machine check verifies exact component coverage, layer order, parent and step agreement, activation and gradient links, output-head dependencies, gradient-norm inputs, global clipping, optimizer settings, and candidate assembly. It also checks the assigned data and signatures. It does not load neural weights or repeat the model computation to accept an ordinary claim.

6.2 Why local replay can expose a global forgery

The transcript fixes all data dependencies. If every local transition is valid on its committed inputs and all edges connect to their prescribed predecessors, their composition is the prescribed graph execution. Conversely, for a forged graph with correctly bound initial inputs, either a structural dependency fails or at least one local transition is invalid. Checking one arbitrary stage does not establish that all stages are valid; an observer must locate an invalid transition or provide the stated coverage guarantee.

The implemented referee loads only one stage's parameters, re-executes its forward pass, its head operation when applicable, backward pass, and optimizer update, and compares the exact committed outputs. Incorrect shapes, nonfinite values, and invalid target masks are execution failures. The first-stage replay includes both tied-weight gradient contributions.

6.3 Availability and disputes

An availability challenger names an input or output object associated with the pending claim. The responsible publisher must place its chunks in native transactions and seal them against the requested digest before the deadline. A missing response rejects the claim. Third parties cannot poison an authorized publisher's upload sequence.

For a computation dispute, the challenger publishes all required replay inputs through finalized native transactions. Only then may the native referee execute. Validators do not fetch arbitrary

off-chain URLs while deciding a block. Missing local durable block data is a node-recovery fault, not a permission to derive a different verdict from network timing.

Uploads are bounded and sequential. False or abandoned computation challenges burn their collateral; returning all of it to a colluding claimant would make repeated self-challenges nearly free. A claim also has an absolute lifetime so challenges cannot lock the single work slot forever. An unfinished fraud accusation at that deadline cancels the claim without paying it, returns publisher collateral, and burns the accuser's bond. A fully elapsed unanswered availability deadline remains a publisher failure. An uncompleted accusation must not itself confiscate an honest publisher's bond. These mechanisms bound specific abuse paths. They do not establish that the chosen token prices fund real network and verification costs.

The terminal replay is deliberately coarse. The two-host dispute published 170,169,856 bytes in 184 upload/seal transactions, requiring 310.05 seconds. Resolution took 10.14 seconds including native transaction processing. This is usable for an experimental objective reference, but expensive for an adversarial public market. Finer authenticated operator transcripts and dispute narrowing should be evaluated against this measured bound; they are not already implemented by calling a stage a succinct proof.

6.4 Total verification cost and sampling limits

The compact check is not the entire verification budget. An honest observer that covers every stage still performs roughly another graph execution outside consensus. If V validators perform compact checks of cost c_s, observers spend c_a per accepted graph, and a fraction q of claims requires a dispute with replay cost c_r and publication cost c_d, an indicative budget is

$$C_{\text{verification}} = Vc_s + c_a + q(Vc_r + c_d). \tag{4}$$

The terms must use consistent resource prices; token amounts alone do not measure CPU, bandwidth, storage, or the opportunity cost of a stalled queue. Our experiments measure components of this expression, not its equilibrium under adversarial demand.

For f invalid stages among S, uniformly sampling k distinct stages after a binding commitment detects at least one with probability

$$p = 1 - \frac{\binom{S-f}{k}}{\binom{S}{k}}. \tag{5}$$

This formula requires the stated sampling conditions and does not establish coverage for the implemented optimistic claims. In a simplified one-shot comparison, if cheating saves compute worth ΔC, forfeits reward R and bond B when detected, its advantage over honest execution is $\Delta C - p(R + B)$. A deterrence inequality therefore depends on detection, actual resource prices and exposure of collateral, not merely on naming a slashing percentage. Collusion, adaptive faults, repeated play and attacks on availability require additional analysis. We do not replace complete observer coverage with a small random sample and then retain the same correctness claim.

6.5 Finality, rewards, and serving state

A disputed invalid transition is rejected before reward issuance. An unchallenged eligible transition settles after its challenge window, relying on the explicit observer assumption. The model transition and rewards commit atomically. Native membership remains independent of which worker computed a gradient; newly issued stake affects voting only through the prescribed bonding and activation rules.

The implemented experiment renews a per-period task budget and issues one experimental token per settled training step. It pays the reserved workers. The funded profile below pays complete-replay services from sponsor escrow, including correct work when no fraud is found. It does not establish a sustainable permissionless audit market. A production policy must fund both audit and availability expenses. Rewarding only fraud discoveries creates the verifier's dilemma. The inflation schedule, long-term fee support, collateral pricing, and initial allocation require public review before an economic launch.

7 Growing the model

For a residual decoder block, write

$$u = h + W_O \ \mathrm{Attn}(\mathrm{RMS}(h)), \tag{6}$$

$$B(h) = u + W_D \left(\mathrm{SiLU}(W_G \,\mathrm{RMS}(u)) \odot W_U \,\mathrm{RMS}(u)\right). \tag{7}$$

The growth transformation copies an existing block's compatible parameters and initializes its attention output projection W_O and MLP down projection W_D to zero. In exact arithmetic, this yields $B(h) = h$. In the implemented floating-point test, finite intermediate activations yield identical tested logits. New output projections receive gradients immediately; inner copied projections can receive gradients once those outputs move away from zero. The added parameters are independent trainable tensors even when initial component bytes coincide.

Four added blocks increase the model to 148,675,392 parameters. The three-worker assignment fails under the declared 48-million-parameter cap, whereas four workers admit it. This demonstrates resource-aware admission and actual parameter growth. It does not establish that either physical host could not hold the whole model under a different allocation.

A growth candidate must preserve old components and tokenizer identity, respect the execution bounds, obtain a feasible resource assignment, train, and pass evaluation before serving. Growth is not a periodic instruction to increase parameter count regardless of value. The evaluation should include a same-training-budget control and measured inference/storage costs. Wider models, expert routing, vocabulary expansion, and a change in numerical profile are distinct transformations requiring their own replay and migration rules.

The native application now admits a separate bonded `grow` claim. Its compact check preserves old tensor roots and architecture fields, restricts added depth, and verifies placement under declared capacities. A growth dispute recomputes the new block bytes using only the last parent block. Successful growth changes the learning head without issuing a training reward or changing the serving alias. A per-period growth budget limits frequency. Subsequent worker sessions begin at the ledger's existing training round, so architecture changes do not reset issuance history. Declared capacities still do not prove that suitable independent workers are online; admission and recovery must make that resource condition operational before public use.

8 Quality promotion and continued operation

The target protocol treats quality evaluation as a separately committed, challengeable computation. A candidate is fixed before the evaluation ticket resolves. The ticket names the baseline, data-selection rule, exposure limits, target construction, metrics, and decision thresholds. Each loss report must link to its model, protected examples, and verifiable forward/head operations. A signed scalar claiming that a model is better is insufficient.

For paired per-document losses, let $d_i = L_i(\theta') - L_i(\theta)$. The research controller computes $\bar{d} + 2.576\, s_d/\sqrt{n}$ as an approximate 99% normal upper bound. Fresh-data improvement must exceed 0.001 nats per scored token; retention allows an upper bound below a 0.02-nat regression margin. At least 32 paired examples are required, and the main experiment uses 64 per group. These choices are explicit research thresholds, not a universal significance or safety guarantee. Dependence, multiple model comparisons, and distribution shift limit their interpretation.

A separate untouched test group is reported after candidate construction. The initial experiment's prompt-dominated metric is retained as a negative methodological result, not repackaged as response improvement. The corrected experiment scores actual response positions on subsequent held-out source rows.

The retention, fresh and test groups here are disjoint hash partitions of the same pinned upstream test split. "Fresh" means previously unused examples in this experiment; it does not establish acquisition of newly published facts or retention across all previous domains. The seed's prior training contamination is not measured by these tests.

An indefinite deployment also needs an error budget across decisions and long-term reference checkpoints. A fixed false-positive allowance reused forever cannot provide a fixed whole-history guarantee. For example, a policy could allocate $\alpha_t = \alpha/[t(t+1)]$ to epoch t and subdivide it among its metrics. If each epoch's tests have the required conditional validity, the union bound gives $\Pr(\text{any false approval}) \le \sum_t \alpha_t = \alpha$. This is a design rule for valid tests, not a guarantee supplied by our small normal-approximation experiment. Similarly, comparing retention only to the last model can accumulate small tolerated regressions; a deployment must also enforce domain-specific floors against durable reference checkpoints. These online statistical and long-term retention policies are not implemented by the current two-group research gate.

The local epoch controller persists data collection, training, candidate commitment, evaluation selection, results, and the accept/reject decision. A rejection keeps the previous accepted model. Restart does not train twice, resample an already reserved evaluation ticket, or consume a second daily epoch allowance for the same active run. This local research registry is not yet the native serving registry.

The opt-in native lifecycle now implements data admission, finite cohort training, evaluation tickets, serving decisions, and paid generation. It requires a fresh experimental genesis and does not itself migrate the public 0.4.0 ledger. Within this profile, subsequent cohorts preserve consensus history, balances, paid-work identities, and the training-round counter. Failed or expired evaluation restores the serving model as the next learning parent.

8.1 Native cohort admission and verifiable metrics

A bonded proposal carries immutable source specifications, consecutive cursor ranges, document identities, provenance commitments, and every assigned token batch. The cohort includes up to 128 training documents and exactly 32 documents in each evaluation group. Each document contains one to four response windows. Training omissions are explicit; evaluation must retain all response targets within the declared policy. Native document and token-batch indexes reject reuse as fresh data. The collector serializes writers before reading a cursor.

Admission requires strictly more than two thirds of both the proposal's snapshotted validator power and the current native power, plus an activation delay. These votes attest to curation, source correspondence, licensing suitability, and tokenization review. They are not cryptographic proofs that data is true, licensed, independent, or useful. Token batches are available in native blocks; raw provenance and model components remain content-addressed replica objects. S3 does not participate in block validity.

A separate reviewer uses its own source and tokenizer policy, reconstructs every response window from the committed raw messages, checks the protected document-hash partition, and rejects heuristic near-duplicates within the cohort. By default it also compares selected messages against pinned upstream Parquet rows, verifying file hashes and repository revision. This check trusts the upstream repository service and does not establish licensing rights or semantic quality. It emits a report rather than an automatic vote; the native application separately checks current cursor, parent, duplicate-history and quorum rules.

Activation waits until the previous cohort closes and no reservation, execution claim, or evaluation remains active. A genesis-fixed budget prescribes one to 64 training steps. Subsequent cohorts schedule a bounded 25% replay fraction from the preceding admitted training cohort when possible. This is a limited replay baseline, not a lifelong representative memory. Each reservation binds the cohort root and exact batch. Exhaustion cannot authorize indefinite mining of the old schedule.

After training, the evaluation ticket fixes candidate, serving baseline, and cohort. A score claim binds up to four assigned windows, with authenticated forward partitions and a separate output-head record. The head is not a trusted scalar reporter: its loss and next-token results can be disputed from the embedding, final norm, and final hidden state. Honest observers still need complete coverage of every accepted graph. Score claims issue no reward. Depending on window counts, a ticket needs 32–128 score claims across the two models and two groups.

The native decision weights window losses by actual target counts within each complete document, using exact rational arithmetic before conversion to a canonical document mean. Each group still has 32 paired observations. Let integer-micronat deltas sum to S, let $Q = n\sum_i d_i^2 - S^2$, and let $G = nm - S$, where m is the permitted margin reduced by a conservative two-micronat allowance. The approximate normal-interval gate is implemented without a floating-point statistical comparison:

$$G > 0, \qquad G^2\,1000^2(n-1) > 2576^2 Q.$$

Both the fresh and retention groups must pass. Public examples, dependence, selection, repeated decisions, and the absence of broad task-level checks still limit what this gate means. It supplies neither a general-quality theorem nor secrecy for the evaluation pool.

9 Inference paid in native tokens

A customer request must bind the finalized serving model, tokenizer, prompt commitment or explicitly public prompt, decoding rule, output limit, provider assignment, expiry, and maximum price. It escrows existing tokens rather than minting a new training reward. A result transcript binds every generated token to the same checkpoint and decoding state. Settlement transfers only the agreed fee, once; expiry returns unused escrow. Model promotion cannot change the checkpoint of an in-flight request.

Paid inference remains deployed in the separate 0.4.0 network. The opt-in evolving-model profile now implements an independent native request/response lifecycle. Requests bind the serving revision, provider, prompt IDs, stop rule, maximum payment, and expiry. Responses commit every autoregressive forward graph, including each output head. Following its dispute window, accepted output pays the provider for actual output tokens and returns unused escrow; unanswered requests refund escrow but not the original transaction fee. A refuted response cannot pay, and an in-flight request retains its original model across promotion.

The current execution profile bounds generation to eight tokens and serializes pending execution claims. It demonstrates a settlement mechanism, not production conversational latency. Model

replication, public provider discovery, versioned recovery, and migration remain separate release obligations. The native experiment is not a deployed evolving paid-inference network.

An assistant resembling a general-purpose personal agent also needs retrieval, user-controlled memory, tools, authorization, privacy, and task-level evaluation. Growing a public LM does not supply those properties by itself. Public blockchain execution does not provide confidential prompts or confidential memories.

10 Experimental evidence

Experiments use two authorized CPU hosts, each with four virtual CPUs and 16 GiB RAM, under one operator. Each guest reports two cores with two threads per core. The processors are Intel Xeon Platinum 8259CL and 8175M, respectively. They test real distributed processes and a native BFT ledger, but not independent economic participation or an adversarial population.

Check	Observed result
Seed implementation vs. pinned reference	Exactly equal tested float32 logits
Identity expansion	134,515,008 to 148,675,392; equal initial tested logits
Distributed whole-model step	Three workers; 8.95 seconds in the conformance run
Compact graph check	0.0246 seconds in that run
Independent stage replay	All three stages reproduced exactly
Native forged-update dispute	Rejected for optimizer-update mismatch after the startup fix
Native valid update	Accepted; exactly 1,000,000 atoms issued
Native state agreement	Matching app hash on all four validators at a common height
Recovery tests	Lost acknowledgment and coordinator/worker restart tests pass
Evaluation correction	Prompt-heavy prefix scoring replaced by explicit response targets

10.1 Response-only quality comparison

Each value is mean teacher-forced response loss in nats per scored token over 64 paired examples. The two response candidates start from the same parent weights and receive the same 32 response-masked batches; the larger candidate first adds four identity-initialized blocks.

Checkpoint	Retention	Fresh	Untouched test
Original seed (135M)	1.325133	1.281919	1.285737
Prefix-trained parent (135M)	1.325136	1.284188	1.285587
Response-trained candidate (135M)	1.324262	1.282754	1.285286
Response-trained candidate (149M)	1.324338	1.282783	1.285116

Neither response candidate passes the predefined promotion gate. Relative to their common parent, fresh-loss changes are -0.001434 and -0.001405 nats, but their approximate 99% upper bounds are +0.000639 and +0.000596; the required bound is below -0.001. The larger candidate has no demonstrated fresh-loss advantage over the smaller candidate under the same training budget.

Relative to the original seed, both candidates also fail the improvement gate. These are negative promotion results, not evidence of continual capability improvement. All candidates produced the same tested 32-token greedy answer prefix to the single solar-panel prompt; that anecdotal output is not a quality benchmark. The chosen thresholds were not relaxed after observing these results.

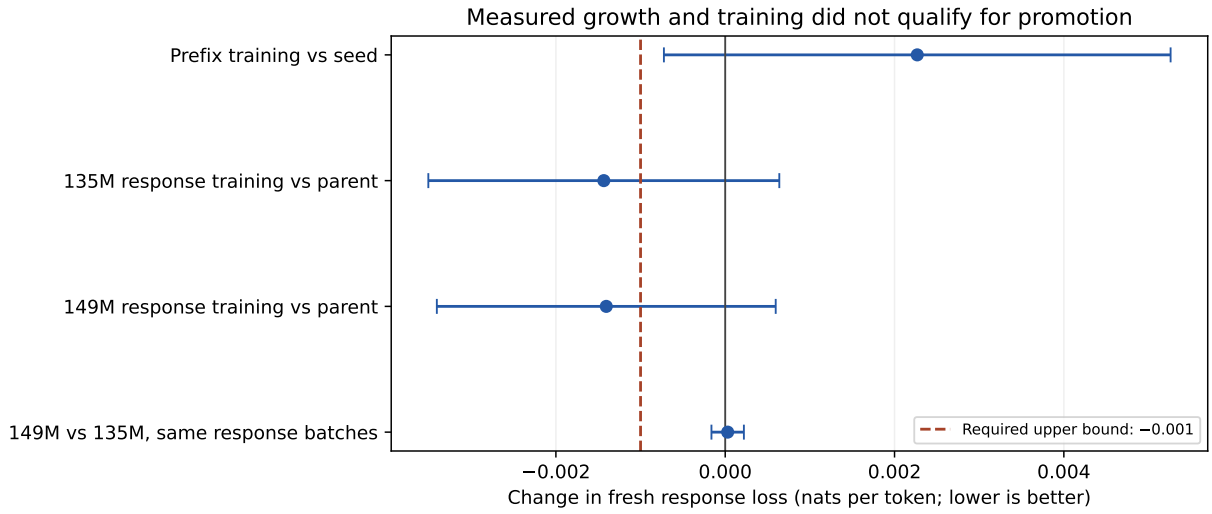


Figure 1: Paired fresh-response loss changes with approximate 99% normal intervals, 64 examples per comparison. Lower is better. Every interval crosses zero; none establishes the required improvement. Intervals are not corrected for multiple comparisons.

10.2 Exploratory response training from the seed

A subsequent plan, recorded before its training began, starts from the original 135M seed and uses 32 batches of two response-masked examples, SGD learning rate 0.03 and the same norm cap. Training uses subsequent source rows 1536–1919; protected examples come from test rows 1536–2303. Each evaluation group contains 128 previously unused paired examples, selected after the candidate commitment. This is an exploratory follow-up motivated by earlier failures, not an independently preregistered confirmatory experiment. The promotion thresholds are unchanged.

Group	Seed	Candidate	Change	99% upper bound
Retention	1.244962	1.249179	+0.004217	+0.011743
Fresh	1.129396	1.138708	+0.009312	+0.017282
Untouched test	1.311664	1.313363	+0.001700	+0.009115

Promotion is rejected: retention remains within the chosen tolerance, but fresh loss increases and fails the required improvement bound. All three stages of the final training step replay exactly. The run therefore provides another concrete separation between correct execution and useful learning. It does not test a larger-model control or establish improved capability. The repository preserves the pre-training plan, source snapshot, candidate, ordered batches, evaluation selection, individual losses and decision, with a bounded reproduction script. None of the quality candidates qualifies for promotion, and no threshold was relaxed to produce a positive result.

10.3 Native architecture transition and resumed training

A further four-validator experiment uses the real model on one physical host. It rejects a forged optimizer update, settles a valid update, rejects a forged identity-growth claim, settles valid growth to 148,675,392 parameters, and settles a subsequent training step from a new worker session at the existing round. The growth referee needs 14,161,224 bytes from one parent block; this smaller figure concerns the expansion transformation, not verification of an ordinary full training stage. Growth itself issues no tokens. The two valid training steps issue exactly 2,000,000 atoms. After restarting the same four nodes, a shared block header at height 514 commits the post-payment application state at height 513. This checks agreement after the final settlement rather than accidentally comparing a header that precedes it.

The timings are measurements under these hosts, code, profiles, and workloads, not throughput forecasts. Concurrent experiments affect wall-clock time. A declared worker parameter budget is distinct from measured process RSS. The single-operator native test does not validate decentralization, an equilibrium incentive mechanism, or public Sybil resistance.

10.4 Text-profile conformance across machines

A subsequent run uses the same 134,515,008-parameter seed and three worker processes across two CPU hosts under one operator. Both hosts independently produce the same codec root and identical token sequences for eight English, Hebrew, Arabic, Chinese, emoji/accent, code, whitespace and mixed-direction probes. All probes round-trip exactly; this establishes text conformance, not multilingual answer quality. The codec preserves the existing 49,152-token vocabulary.

A manually written two-answer fixture contains 121 response targets in three windows, compared with 64 retained by the legacy first-response preparation. The actual training batch contains only the first two windows, totaling 101 targets. One full-model update takes 18.05 seconds; the three stage replays pass in 7.32, 6.41 and 5.76 seconds. Eight generated tokens decode to "The capital of France is Paris." The updated model and a 148,675,392-parameter growth candidate retain the codec identity. This is a functionality check after one update of a pretrained model, not evidence of improved capability. The public source includes the text conformance script and its bounded runtime profile.

An initial native run of this text-bound record stopped during dispute resolution when the synchronous RPC response expired while the referee was executing. Transport failure did not establish transaction rejection. The experiment driver was corrected to submit a signed envelope once and query its exact hash for a finalized outcome, with an explicit confirmation deadline. Lost acknowledgments, final rejection and an unresolved deadline are separately tested. This failure also emphasizes that the operational cost of a referee includes its effect on transaction and consensus latency.

The corrected run completes the real-model lifecycle with four validators on one host under one operator. It rejects forged training, pays valid training, rejects forged growth, accepts growth, and pays the following training step. Training replay publishes 171,114,496 bytes through 184 upload/seal transactions in 363.57 seconds; resolution takes 19.47 seconds. Growth replay publishes 14,161,224 bytes. The two training steps issue 2,000,000 atoms in total; growth issues none. All validators agree on the application hash in block 539, which commits the final paid transition at height 538. Tokenizer identity is preserved, while the serving head remains on the original model. This is an isolated settlement result, not a demonstration of independent ownership, economical verification or improved serving quality.

10.5 Integrated data admission and serving checks

The new lifecycle is first tested with a 10,384-parameter synthetic model, four validators on one host, and one operator. Two cohorts activate without resetting the chain. Four training settlements issue exactly 4,000,000 atoms. The observer independently checks 107 stages. A synthetic quality gate passes, a forged output head is refuted using 4,520 bytes of replay evidence, and valid generation transfers 1,000 atoms while refunding 6,000. These are protocol tests, not evidence of LLM improvement. With one validator stopped, blocks continue; with half the voting power stopped, finality halts. Restarting the same nodes restores progress and agreement at a common block.

For the real-data trial, two further admitted cohorts use the unchanged SmolLM2 text contract and pinned Smol-SmolTalk source. Each contains 32 training, 32 retention and 32 fresh-evaluation documents. The first has 117, 69 and 75 response windows respectively; the corrected second cohort has 111, 89 and 76. The separate reviewer reconstructs every window and checks all 192 original documents against their pinned upstream rows. The first training interval is rows 896–1,152 and the second 1,152–1,664; held-out intervals are 0–384 and 384–896, with interval ends excluded. This means fresh source rows, not newly acquired world knowledge or independent curation. A one-window preparation policy rejected too many complete multi-turn responses and was replaced by the bounded multi-window format. Longer excluded responses remain a source of selection bias.

The first real trial prescribes four training steps drawn from 117 admitted training windows. Admission is not exhaustive training. The following cohort's replay pool includes previously admitted windows that may not have been selected earlier; it is not necessarily rehearsal of previously performed updates. Both evaluation groups also derive from the same dataset family. These policies need data-utilization, memory and cross-domain studies before any claim of lifelong learning.

The real native trial expands the 134,515,008-parameter seed to 148,675,392 parameters and settles four full-model training steps, issuing 4,000,000 experimental atoms. Growth issues none. Evaluation settles all 288 assigned windows through 74 native score claims. Retention mean response loss changes from 1.429667 to 1.429275 nats; its conservative upper bound on the change is $+0.000311$, below the allowed $+0.02$. Fresh mean loss changes from 1.307629 to 1.306425, but its upper bound is -0.000322, which does not meet the required bound below -0.001. The native decision at height 2,865 therefore rejects promotion. The original serving model is preserved and restored as the next learning parent, while paid computation and candidate measurements remain in history. This is a useful protection result and a small observed loss reduction, not a demonstration of sustained model improvement.

The native inference dispute then publishes 113,334,000 bytes of output-head evidence and rejects a deliberately forged next token. Valid one-token generation pays 1,000 atoms and refunds 6,000. The completed original driver path covers 353 independent stage checks, reconstructed from its finalized honest claims and completed checkpoints after the subsequent interruption.

The first attempt to admit the second cohort fails: two different source documents contain token windows already admitted in the first cohort. The native freshness guard rejects the proposal. Individual provenance checks and document-ID deduplication were insufficient. The preparer and reviewer are extended to exclude identities and token windows from retained prior admissions. The corrected proposal selects replacement documents without collecting additional source rows. Recovery reopens the same genesis, signing keys, blocks and application databases, preserving all completed payments. After corrected admission, a one-validator outage permits progress, half-power outage halts finality, and restart restores agreement. A non-voting full node on the second host replays the same chain, including the dispute, and matches the application hash in block 3,163. Its peer reconnect backoff requires an operational restart after the interruption; this is not an uninterrupted availability result.

A further continuation trains the corrected second cohort using three fresh windows and one window from the preceding admitted pool. Twelve more stage checks pass. The preserved ledger advances from round four to round eight and total issuance from 4,000,000 to 8,000,000 atoms in 185.73 seconds. These updates start from the original 134,515,008-parameter model after the earlier rejection. The serving root stays unchanged pending the second evaluation, which this bounded run does not perform. All four validators and the second-host follower agree in block 3,346. The isolated test processes are then stopped with their state retained. This verifies continued paid work after fresh-data admission and recovery; it is not yet an unattended public continual-learning service.

11 Compact optimizer refutations

11.1 Committed assertions and their limits

The opt-in model field `update_witnesses` selects profile `neuroshard-sgd-chunks-v1`. Each training stage commits a Merkle root and exact shape for every owned parameter before the update, its asserted gradient, and the parameter after the update. The existing numerical operation is unchanged. The declarations assert both correspondence with the flat SHA-256 safetensors objects and the prescribed forward/backward computation.

For one tensor, denote those assertions by A_1 (parent correspondence), A_2 (correct gradient), A_3 (candidate correspondence), and A_4 (correct SGD relation). A publisher claims $A_1 \wedge A_2 \wedge A_3 \wedge A_4$. Authenticated chunks contradicting A_4 refute this conjunction without first proving the other assertions. Agreement on A_4, conversely, proves none of A_1, A_2, A_3. These indexes are optimistic declarations, not cryptographic subfile proofs for a flat SHA-256 digest. A producer can invent a zero gradient and unchanged output satisfying A_4; a regression test accepts that compact relation while the full-stage oracle rejects the training.

Tensors use contiguous row-major little-endian float32 bytes and chunks of 1,024 elements. Leaf hashes bind a domain, position, byte length and contents; internal nodes have another domain, with canonical empty padding to a power of two. The metadata binds exact integer shapes, names, order and complete stage ownership. A witness opens before, gradient and after chunks at the same position. Its validator executes the pinned in-place float32 operation `before.add_(gradient, alpha=-rate*scale)` and compares bytes exactly. No epsilon or probabilistic spot-check changes acceptance.

The native `refute_update` transaction carries at most 32 KiB of witness data. It reads no off-chain artifacts and invokes no full neural referee. Invalid paths reject the transaction without changing balances. Valid openings of a consistent relation burn the accuser's bond without extending the claim deadline. An inconsistent relation rejects the claim, returns the challenger bond, pays half the publisher collateral to the challenger and burns the remainder. The rejected task issues no tokens. Compact checks are counted separately from complete referee calls.

11.2 Cost and reproducibility

Constructing the witness still requires the observer to reconstruct the gradient and check its relationship to the declared index. Incorrect earlier operations, false indexes, unavailable bytes and coherent forged gradients retain the full-stage fallback. This is an improvement to one adjudication path, not a demonstrated reduction of all observer work or worst-case adversarial cost. The complete target remains a committed forward/backward graph with bounded, reproducible disputes for every supported operation.

The benchmark alternates unchanged and instrumented executions of the same 134,515,008-parameter numerical task, reports the first cache-warming pair separately, retrieves and hashes output components, and checks identical parameter bytes, loss and payable work identity. Repeated timing samples are not fresh training or separately payable tasks. It measures complete stage audits, witness preparation, compact checking and actual native transport/finality costs for the same forged update. A second CPU checks portable witnesses and 45 fixed numerical stress cases covering tails, signed zero and subnormals. Finite conformance testing does not justify extending the numerical profile to arbitrary hardware.

The warm training medians are 16.167 seconds without commitments and 22.209 seconds with them, a 37.4% increase over three samples per variant. Claim metadata grows from 16,383 to 116,948 bytes. Complete audits of all three stages take 15.565 and 22.640 seconds, respectively. These are timings for repeated identical tasks with cached artifacts, not sustained throughput for different batches and newly changing checkpoints. The first unchanged run sends 539,868,797 object bytes and retrieves 539,861,200; the following instrumented run reuses those artifacts and transfers only new metadata. Warm samples perform no new object-payload transfer.

The forged stage requires 184,644,776 input bytes for complete replay and takes 7.316 seconds on the first CPU. Its compact witness is 19,521 bytes, takes 6.103 seconds to construct from cached artifacts, and checks in a median of 1.235 milliseconds on the first CPU and 0.957 milliseconds on the second. The second CPU also accepts the honest stage through the full oracle in 7.768 seconds and rejects the forgery in 7.021 seconds. Both CPUs produce the same digest for the 45 numerical stress cases.

The evidence reduction applies to the tested inconsistent optimizer relation. An adversary may instead publish a coherent invented gradient that still requires the complete dispute path. The added honest-work cost and remaining observer work preclude a claim of economical verification overall.

In the separate native test, four validators on one host reject the same forged record through both paths. The complete dispute submits 193 signed transaction envelopes totaling 246,290,862 bytes and takes 428.132 seconds; its final resolution transaction takes 22.412 seconds. The compact dispute uses one 19,972-byte signed envelope and finalizes in 0.848 seconds. Common reservation and claim setup is excluded from both measurements. RPC framing, consensus gossip and repeated peer transmissions are also excluded: these are submitted envelope sizes, not aggregate network traffic. Both paths run in the same sequential native profile. The resulting comparison does not establish a production latency, general verification speedup or an independent compute market.

The false compact accusation burns its bond without extending the deadline. The correct task issues exactly 1,000,000 experimental atoms once; all four validators agree on the post-payment state committed in block 458. The serving checkpoint remains the initial model. The test records one full referee call and two compact checks separately, and stops all its processes. Source commit `3bf6a08` and the checksummed numerical, native and second-CPU evidence are published at https://neuroshard.com/experiments/update-witness-20260911/SHA256SUMS. These new tests do not upgrade the public 0.4.0 network.

12 Funded complete-replay integration candidate

12.1 Purchased obligations and acceptance

An additional opt-in genesis profile implements prepaid, collateralized auditing for training, growth, evaluation and generation. A sponsor selects one to eight auditor accounts and locks

$$B_{\text{audit}} = p\, s_{\max}\, k,$$

where p is the fixed price per execution stage, $s_{\max}$ is the purchased stage bound and k is the auditor count. Each selected auditor accepts the obligation and locks its own bond before a training reservation or direct execution claim can attach that budget. The publisher and reserved worker keys must differ from the auditor keys. This is an account-separation rule, not evidence of different ownership. The sponsor chooses the committee; permissionless, collusion-resistant selection remains unimplemented.

The claim's committed execution graph determines its actual stage count. Insufficient coverage rejects the claim. Every selected auditor must cover the complete graph, including generation's output heads. The implemented daemon fetches and hashes the required inputs and claimed outputs, replays every stage through the existing referee, and stores its result and salt durably before sending a commitment. Missing data leads to an availability request; an incorrect execution leads to the native uploaded-input dispute.

Reports use separate commitment and reveal windows. The digest binds the chain, claim, auditor account, complete-coverage root and salt. All early commitments can end the commitment phase; reveals begin in a subsequent block. A complete set of reveals starts a full ordinary challenge window. The coverage digest is public and cheap to compute. It specifies the purchased obligation and does not prove that replay occurred. Private result sharing and colluding false attestations are still possible.

A claim with missing reports cannot settle or mint. On acceptance, each auditor receives $p\, s_{\text{actual}}$ from the sponsor's existing tokens, its bond returns, and the unused stage allowance refunds. Training keeps its existing one-payment-per-numerical-work identity. Audits, growth and scoring do not introduce additional training issuance. Reserved service funding cannot be cancelled while execution is pending.

12.2 Failure accounting

An objective fraud verdict cancels the service payment and slashes revealed false attestations, dividing collateral between the successful challenger and burn. An auditor that refuses to attest to proven fraud is not punished for that refusal. Missing reports incur a timeliness penalty only after an uninterrupted reporting window. Publisher withholding or an unresolved allegation does not prove that an auditor shirked. An intervening native availability or full fraud challenge therefore exempts this timeliness penalty, while leaving objective false-report liability intact.

This conservative exception admits a bonded griefing strategy: interrupting the window can avoid a missed-report penalty and prevent progress. Existing challenge collateral and absolute expiry bound one episode, but do not establish favorable repeated-game incentives. Requiring all selected reports also makes liveness dependent on every selected auditor. The candidate implements purchased coverage and accountable reports under explicit observer assumptions, rather than resolving the verifier's dilemma.

The prototype also caps the global pool at sixteen outstanding audit offers. A funded sender can fill it with unaccepted offers whose escrow is refundable, blocking other sponsors until cancellation or expiry. The permitted lifetime is up to 100,000 blocks. An adversarial test records this admission denial of service. A larger fixed pool would trade the same issue for greater state use; resource-priced admission remains a release requirement.

The ledger includes sponsor funds and auditor bonds in the invariant

$$S_{\text{initial}} + S_{\text{issued}} = S_{\text{liquid}} + S_{\text{validator}} + S_{\text{escrow}} + S_{\text{burned}}.$$

Cancellation before reservation, unused offers and expired reservations return unearned service funds. Issuance remains the training-round count times its fixed reward. A restart or a repeated signed

submission cannot authorize another payment for the same work.

12.3 Recovery and service costs

A continuous operator funds the next audit, reserves an admitted training batch, recovers durable worker operations, publishes artifacts, submits its claim, and completes the quality evaluation. When that work ends it can serve addressed inference or wait for another natively admitted cohort. It does not automatically admit arbitrary collected data or grow the model without another decision. Training and evaluation occupy a serial slot; inference can expire while they run. This is an integration scheduler, not a low-latency serving result.

Both publisher and auditor use a SQLite outbox that stores the intent and exact signed envelope before broadcast. Recovery first looks up CometBFT's SHA-256 of those signed bytes and otherwise retransmits the same envelope. The application's logical claim identifier excludes the ECDSA signature and is a different digest. An unresolved outcome blocks signing another nonce. A permanently rejected operation needs investigation, not deletion of the outbox. Source and genesis commitments are checked before participating; same-source crash recovery does not demonstrate a consensus upgrade.

With the reference audit price, four partitions and one auditor, a response ending after one token incurs 500,000 atoms of audit expense plus two 1,000-atom publisher transaction fees. The experiment's 1,000-atom token price is consequently subsidized. The current operator refuses such jobs unless its configuration explicitly enables a subsidy. It also defaults to a one-token response bound within the native eight-token maximum; larger configured responses require matching auditor capacity and measured execution times within the genesis reporting windows. That price floor does not include compute, artifact retention or a margin. Funding is per claim, not a prefunded guarantee that an entire future cohort, evaluation and serving period can finish. Sustainable prices and retention liabilities remain separate requirements.

Audit acceptance currently locks no separate budget for refutation. An adversarial test gives an auditor sufficient funds for its audit bond but not for the additional challenge collateral. It detects a forged update, cannot open the dispute and declines to attest. The claim cannot mint, but the uninterrupted missed-report rule burns the auditor's bond. The real trial avoids this condition by initially funding its auditor with 100,000,000 atoms. Its measured head dispute needs 2,000,000 atoms of challenge collateral and 116,000 atoms in dispute fees, in addition to the 5,000,000 audit bond and acceptance fee. Prepaid refutation and completion capacity remain a protocol requirement beyond the current purchased-replay service.

12.4 Starting with one operator

A single owner can start a public experimental chain and test execution, accounting, failures and joining. That is a bootstrap arrangement, not an independent quorum. A new participant can first follow the agreed genesis with its own non-voting full node, then accept funded work with its own account. Independent consensus requires distribution of voting power and key custody, with the hosting and ownership topology disclosed. Increasing the number of processes under the original owner cannot satisfy that requirement. No outside participants are claimed in the experiments below.

12.5 Funded candidate measurements

The new trials freeze the scripts at commit `3710226` and package source commitment:
`2875b6e2de443c136cfffe3288b35d03a3f763ee521ce4f231d6704043e1a125`

Four validators run on the first CPU host. The real-model pipeline places two workers there and two on the second host, which also runs the funded auditor and a non-voting full node. The hosts are Intel Xeon Platinum 8259CL and 8175M machines, each with four logical CPUs and approximately 16 GiB RAM. All belong to one operator. The public 0.4.0 services also use these hosts; these measurements are not isolated performance benchmarks.

The synthetic rehearsal completes eight training settlements over two cohorts, including a coordinator kill after reservation and successful recovery through the second evaluation. It issues 8,000,000 atoms for eight distinct tasks and pays 21,100,000 existing atoms for 73 audit services. The auditor records 214 stage checks, of which three reach the deliberately forged head. Its native refutation uploads 4,520 bytes. Honest one-token inference pays 1,000 atoms and refunds 6,000. Both synthetic quality gates pass; this fixture does not establish LLM improvement. One validator's loss permits progress, half the voting power halts, and restarting the retained nodes restores agreement.

The real trial repeats identity growth to 148,675,392 parameters and settles four full-model updates before its first quality decision. Growth creates no issuance. The 32-document retention group passes its tolerance, but the fresh group fails to establish the required 0.001-nat improvement. The original 134,515,008-parameter checkpoint remains the serving and next learning parent. This uses the earlier reviewed Smol-SmolTalk cohorts and fixed text contract; previously unused source rows do not establish new world knowledge.

The second cohort completes four more training settlements and evaluates a 134,515,008-parameter candidate. Its retention group passes, but its fresh group also fails the improvement gate. Quantized mean document-loss changes for the first and second candidates are respectively -0.00120472 and $+0.00002372$ nats on fresh responses. The first point estimate improves, but its uncertainty bound does not establish the required improvement. Both decisions retain the original serving root. Across both cohorts the ledger issues exactly 8,000,000 atoms for eight distinct numerical tasks, and transfers 70,200,000 existing atoms for 168 accepted audit services. The auditor records 169 reports and 706 stage checks: 702 stages belong to paid services, and four checks reach the forged response head.

The coordinator automatically recovers its second-cohort reservation after being killed. The frozen experiment driver later exceeds its one-hour wait while scoring remains unfinished and stops its child services. A manual, bounded continuation resumes the same homes, keys, signing state, workers, auditor and transaction outboxes at height 8,968. It completes the remaining 68 candidate fresh-response windows without additional training or a new genesis. The evidence distinguishes this intervention from automatic transaction recovery. The current driver exposes a configurable two-hour wait; this does not extend ledger deadlines.

The real auditor detects a forged output head after four checks in about 5.79 seconds, including object handling. During its native fraud dispute, the failure injector pauses the process, confirms a signed upload has no local receipt, and kills it. Automatic restart retains the exact envelope and nonce 244 and obtains its finalized receipt. A missing local receipt does not imply that the transaction was not already on the chain. The recovered auditor finishes the 113,334,000-byte dispute, and native replay rejects the forged head at block 5,265. The 116 unique signed transaction envelopes total 151,171,234 bytes. Their first and last consensus block timestamps span 342.559577 seconds, including the injected crash. This excludes retries, RPC framing and peer gossip; it is not a monotonic client-latency measurement or aggregate bandwidth. This is recovery of a substantial evidence transfer, not a succinct proof. Honest generation then pays 1,000 atoms and refunds 6,000. The retained serving model uses three partitions, so its four audited forward stages cost 400,000 atoms before the two publisher submission fees. This response is subsidized.

The second-host full node preserves its keys and genesis while moving from an SSH peer tunnel to the public native endpoint. With the peer tunnel removed, all five full nodes agree at block 3,588 on application hash `AA3A1C5372395FB98A70C3857D8C41B58A57C8E0802499A426BB662737DF3FE3`. Pub-

lic connectivity and cross-host replay do not establish independent voting power or tolerance of losing the primary host.

After the long driver outage, a manual restart of the retained follower restores public-peer agreement at block 9,086. Following the completed evaluation, three of four validators continue; half the voting power halts finality; restarting the retained validators restores agreement in block 9,811. All four final databases satisfy the supply invariant: 3,997,019,000 liquid atoms, 10,000,000 validator collateral, zero escrow and 10,981,000 burned atoms equal the 4,010,000,000 initial atoms plus 8,000,000 issued atoms. Both bounded trial networks are stopped with state retained. The public 0.4.0 network continues separately.

The complete genesis, results, replay reports, accounting and recovery records are published with SHA-256 checksums at the funded-audit evidence archive. These are inspectable measurements by one owner, not proof of independent participation or a generally improved language model.

13 A preregistered useful-learning milestone

The preceding integration experiments establish executable transitions but reject their quality candidates. A subsequent experiment freezes the recipe, implementation, document identities and generation prompts before training. Its first question is whether additional full-model training produces a measured improvement on this declared distribution. Continual learning and two-host reliability are subsequent, conditional experiments; neither can rescue a failed first phase.

13.1 Commitment and preparation

The experiment starts from the pinned 134,515,008-parameter SmolLM2-135M-Instruct seed, bound to the versioned text codec. The pinned Smol-SmolTalk source revision is:

`f73fe857d519ff6ac5af2ea67c4d3834da7b8bcc`

The training scan covers rows $[8192, 10240)$ of its training split; the held-out scan covers rows $[3072, 5120)$ of its test split. The entire declared ranges are scanned. Selection preserves 256 complete training documents and 64 documents in each held-out role: retention, fresh and test. Document identities assign held-out roles, and duplicate exclusion precedes the fixed selection order. No loss or generation is used for selection.

The committed manifest identifies all 448 documents, source positions, window hashes, training batches and generation inputs. It binds the original plan and a digest of the numerical implementation and dependency profile. Training rejects uncommitted or changed plan/selection bytes. The executable revision is `4f18f903b738d5f2e5ba6d31434b7e0ae5e06fd7`; the complete manifest and roots are available with the public evidence. A reconstruction in a fresh local directory reproduces the exact selection bytes. This reconstruction has the same host and operator; it is not an independent-ownership experiment.

"Sealed" means committed before this run, not secret. The upstream seed already underwent Smol-SmolTalk fine-tuning. This experiment therefore does not establish that its evaluation documents were absent from the seed's training history. It measures additional training against that same seed.

13.2 Fixed training and evaluation

One physical host runs three workers with declared capacities of 48M parameters each. There is no growth, second host, replay or learning-rate search. Each selected training document supplies one hash-selected response window. A hash order pairs these into 128 batches of two windows. SGD

uses learning rate 0.003 and global clipping at 1.0. Unused admitted training windows are retained but do not count as trained rehearsal data.

Both models score all selected evaluation windows: 149 retention, 145 fresh and 162 test windows. Losses cover assistant-response targets only. Each document contributes one observation, obtained by weighting its window losses by actual response-target counts. The paired rule uses $\bar{\Delta} + 2.576 s_\Delta / \sqrt{64}$, with candidate-minus-parent loss. Pass requires the sealed test upper bound below -0.001 nats and the retention bound below $+0.02$. Fresh results are reported but cannot rescue or veto that decision. These approximate normal intervals are conditional on the selected evaluation distribution, not a broad capability guarantee.

The first 20 sealed test documents in sorted identity order supply their last user turns as generation prompts. Both models receive the same chat rendering and greedy decoding for at most 32 new tokens, stopping at EOS. The existing executor retains the last 192 rendered prompt tokens; seven of the 20 prompts need this truncation. Original prompts, token counts and truncation flags remain in the evidence. Every response is published, including empty or poor outputs; generations do not change the loss gate.

The experiment has a 72-hour wall-clock budget, including evaluation, retries and downtime, and a 256 GiB artifact budget. The retained coordinator journal and worker receipts bind accepted steps and work identities across restarts. No public token is issued and the public 0.4.0 network is unchanged.

13.3 Observed result: reject

All 128 prescribed training steps completed, with 128 distinct work identities, 384 worker receipts and no transport retries. The sum of observed training-step times was 2,309.85 seconds, with a median of 18.27 seconds. The complete phase took 7,625.03 seconds (127.08 minutes), including scoring and generation, and retained 66.92 GiB within the declared disk budget. Input reconstruction overlapped part of training, and other public services remained active. These timings are not an isolated benchmark.

Role	Parent	Candidate	Mean change	Upper bound
Sealed test	1.420849	1.410772	−0.010078	+0.004007
Retention	1.497543	1.487507	−0.010036	+0.002262
Fresh (reported)	1.551739	1.541011	−0.010728	−0.004949

The sealed-test upper bound is above the required -0.001, so the phase fails. Retention passes its $+0.02$ bound. Fresh also improves at its reference margin, but cannot rescue the sealed test. The candidate is retained as an openly downloadable research artifact and is not promoted on the public chain. The bound candidate root begins `ecb4262c2b7d`; the complete root, selection digest and archive hash are in the evidence.

Post-scoring inspection finds 41 test documents improved and 23 worsened, with a median change of -0.001634 nats. A document with only two response targets improved by -0.333457 nats, contributing strongly to the mean and its uncertainty. It remains in the evaluation. This heterogeneous result cannot be reclassified by deleting an outlier, changing aggregation or using a different significance test after inspecting the scores.

A post-run diagnostic rescored the largest test gain and largest retention regression one target at a time. In the former, loss on `False` decreased by 0.60956 nats and end-token loss by 0.05735. In the latter, loss on the content token for “three” increased by 0.84699 despite an end-token improvement of 0.16558. Changes therefore involve answer content in both directions. The 12 read-only forwards

took 65.28 seconds after the primary run and reconstructed its window means within 10^{-6}. This diagnostic uses two documents selected after scoring and defines no new quality test.

Eleven of the 20 response-token sequences changed and nine were identical. Fifteen responses from each model used all 32 permitted output tokens. Some repetition decreased, but an elementary earthworm question retained its incorrect answer and several code examples remained unfinished. These probes do not establish a better general assistant. All responses and original/rendered inputs are published.

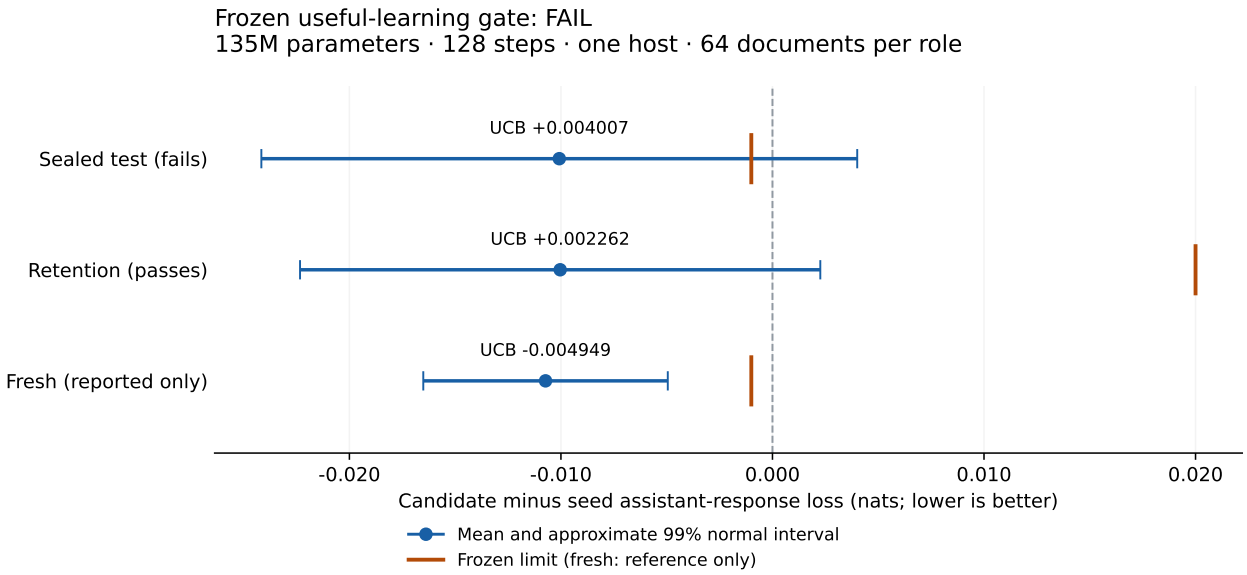


Figure 2: Paired document-loss changes in the frozen 128-step experiment. Intervals are the preregistered approximate normal intervals; acceptance uses their upper endpoints. Fresh is reporting-only.

An auxiliary, read-only lossless-storage probe encoded the final transition's 32 tensor components as bytewise XOR against the exact parent, followed by zlib compression. The 538,085,392 raw bytes became 153,503,190 bytes; direct compression of the candidate took 422,376,714 bytes. Restoration reproduced every target byte and component hash. Delta encoding and restoration/hash verification took 9.83 and 4.80 seconds in one shared-host observation. This is a possible transfer or archival encoding when the parent is present, not a neural-work proof, network-throughput result or authorization to prune active dispute evidence.

13.4 Conditional next phases

Only a phase-one pass permits the second cohort. It must use subsequent unused source ranges and preserve the first sealed test as an untrained retention probe. One quarter of its schedule rehearses windows actually trained in phase one. A pass requires a new sealed-test gain relative to the phase-one candidate and retention of its earlier test performance. Even a pass in both phases would concern two slices of the same corpus, not lifelong learning or a new domain.

The two-host phase separately compares identical 16-step schedules on one host and two hosts. It must kill a worker after step eight, recover the accepted journal without duplicating work identity or payment, and finish with the same model root. It reports transfer, idle time, retries and wall-clock overhead. At 135M this is a test of an additional failure domain; it is not a claim that either machine

lacks capacity for the model, that throughput increases, or that an independent operator has joined.

14 Conventional GPU learning reference

The failed 135M experiment combines a small seed, short response windows, plain SGD and the distributed execution implementation. To isolate a conventional baseline before protocol integration, we conduct an operated full-model experiment with the Apache-2.0 SmolLM2-1.7B-Instruct seed at revision `31b70e2e869a7173562077fd711b654946d38674`. Importing this larger upstream model is not growth produced by NeuroShard. The run makes no native transition, pays no tokens and does not change the serving head.

The committed preparation contains 2,048 complete training conversations from pinned Smol-SmolTalk, with 128 development, 128 retention and 256 final-test documents. Conversations longer than 1,024 tokens are excluded. All real assistant tokens, including EOS, are supervised; prompts and control tokens are masked. The seed already had related SmolTalk training. Disjoint partitions within this experiment do not establish unseen upstream data.

All 1,711,376,384 parameters train for 256 AdamW updates, eight documents per effective batch, totaling 655,372 assistant targets. Parameters, gradients and Adam moments are FP32; CUDA operations use BF16 autocast. Peak learning rate is 3×10^{-5}, with 16 warmup steps, cosine decay to ten percent, weight decay 0.01 and global gradient clipping at one. The final step is selected in advance. Its exact checkpoint is committed in Git before the final test is opened; no intermediate checkpoint is selected by test score.

Role	Documents	Seed loss	Change	Approximate paired 99% interval
Development	128	0.580172	−0.008030	[−0.015449, −0.000612]
Retention	128	0.544322	−0.003980	[−0.012793, +0.004833]
Final test	256	0.536945	−0.003248	[−0.010635, +0.004139]

Table 1: One GPU reference run. Loss is assistant-response nats per target token, averaged once per document; change is candidate minus seed. Intervals use the paired standard error and multiplier 2.576. They do not measure uncertainty across training seeds or certify broad assistant quality.

The final-test interval includes zero. Twenty generation pairs per role, using the complete prompts and a shared 256-token greedy limit, receive an unblinded development-agent review under a rubric committed before training. Development preferences are candidate 1, seed 5, tie 11, neither 1 and unresolved 2; final-test preferences are candidate 4, seed 2, tie 9 and neither 5. These judgments are not an independent human evaluation. Test token-limit exhaustion falls from eight seed answers to five candidate answers, but correctness remains mixed: a deductive-reasoning example improves while a Pandas string-conversion answer regresses. Neither model correctly implements all requested feature-extraction edge cases. All answers, judgments and executable counterexamples are published.

One L40S on a g6e.2xlarge completes the development driver in 1,257.38 seconds; the complete process takes 21 minutes 15.14 seconds. The 256 update timers sum to 437.94 seconds. Peak allocated Torch CUDA memory is 31.599 GB and peak retained artifacts are 82.184 GB. The separate final-test process takes 7 minutes 35.95 seconds, including cold seed-file verification. Restoring the step-192 Adam/RNG checkpoint and repeating 64 updates takes 170.19 seconds and reproduces every numerical journal field and all 218 final parameter tensors bit for bit. This is a same-host recovery check; it establishes neither cross-GPU agreement nor cheap adversarial verification. The GPU was bounded by a $100 budget and a twelve-hour automatic stop deadline, then terminated after evidence preservation. Its public receipt accounts for the observed compute lifetime at $2.24208 per

hour, excluding storage, traffic and taxes. The downloadable model archive preserves exact FP32 weights and tokenizer files; optimizer state requires the retained operator checkpoint or reproduction of the public recipe.

Reference quality is itself a limitation. A development target uses "happy" once despite requiring it three times. A new report-only parser checks narrow instruction templates: after fixing a placeholder-induced false positive, it checks 91 constraints in 81 training records and flags four records. These are contextual review findings, not a general quality estimate or an admission rule. The next experiment must freeze fresh evaluation data and compare reviewed targets with an unchanged-seed and unreviewed-data baseline at equal budget. Lower loss against imperfect targets cannot substitute for useful answers. The prior frozen contract remains failed, and no continual-learning or scaling gate is unlocked.

The reproduction code, input and candidate commitments are in the public repository. Complete outputs, runtime details, checksums, model weights and recovery evidence are at https://neuroshard.com/experiments/learning-reference-gpu-20260913/.

15 Measured learning and cooperation on two GPUs

The next operated experiment separates checked target quality, synchronous training cooperation and inference replication. It uses the same pinned 1.7B seed, with 768 generated grounded training tasks and 256 public conversation-replay records. Four task families require strict JSON answers: record lookup, status filtering, priority sorting and integer invoice totals. Targets are computed from the supplied records before tokenization. A control deliberately damages 203 of the 768 task targets while retaining the same prompts, conversation replay and document schedule. This artificial corruption is not an estimate of errors in an upstream corpus.

All three arms train every parameter for 128 AdamW updates at a global batch of eight. Grounded response tokens receive weight eight and replay tokens weight one, chosen from target counts before scoring. The clean arm has 84,133 unweighted targets and 150,514 weighted targets; the damaged strings produce 84,660 and 154,730, respectively. Evaluation remains unweighted. Each clean-pair rank processes four documents. If S_r is its weighted summed loss, D the global weighted target count and $W = 2$ the world size, the local backward objective is WS_r/D. DDP averaging therefore produces $\sum_r \nabla S_r/D$, preserving the intended global objective despite unequal response lengths. Parameters, gradients and Adam moments remain FP32, with BF16 forward/backward autocast.

The preparation is committed before training, which runs at `a7efc6c`. All three predetermined step-128 candidates are committed together at `394fb106` before the final test. The shared checkpoint is selected for the serving experiment before test outcomes are available. No candidate is chosen from intermediate checkpoints by final-test score.

Final-test family	Seed	Clean, one GPU	Damaged, one GPU	Clean, two GPUs
Lookup (64)	23	62	60	63
Status filter (64)	0	60	47	58
Priority sort (64)	0	47	45	47
Invoice totals (64)	0	0	0	0
Total (256)	23	169	152	168

Table 2: Every response is checked for exact schema and values. All answers and failures are published. These are four generated task families, not a broad assistant benchmark.

The primary clean-single arm gains 146 correct answers and loses none of the seed's correct

answers, an accuracy gain of 57.03 percentage points. The exact one-sided paired McNemar test gives $p = 2^{-146} \approx 1.12 \times 10^{-44}$. Mean response-loss change on 128 public retention probes is -0.002195 nats, with an approximate paired 99% upper bound of $+0.006656$. The predefined thresholds are a gain of at least five percentage points, $p \leq 0.01$ and a retention upper bound at most $+0.02$. This narrow contract passes. The test is conditional on these examples and this trained seed; it does not measure uncertainty across training runs.

Test examples contain fresh generated records, and half use two wording variants absent from training. Clean-single answers 83/128 of those correctly, compared with 9/128 for the seed. The four task definitions remain shared with training. Valid JSON objects increase from 198/256 to 256/256; schema compliance accounts for part of the gain. The arithmetic failure persists: for an invoice totaling 365, both clean candidates answer 431. Every model fails all 64 totals, so the recipe does not establish reliable arithmetic.

Clean-single gains 27 and loses 10 answers versus the damaged control, a net 17/256; this comparison is descriptive, without multiplicity or multi-seed validation. The paired candidate gains three and loses four answers relative to clean-single, within the predefined two-percentage-point descriptive margin. The retention probes were previously exposed, and the seed had related upstream SmolTalk training. Neither those probes nor this generated test certify general retention. The earlier failed 135M contract and inconclusive general 1.7B reference retain their original outcomes.

Measurement	Clean, one GPU	Damaged, one GPU	Clean, two GPUs
Update timers (s)	208.70	209.55	917.03
Training loop (s)	986.84	665.01	1,294.04
Allocated GPU loop time (s)	986.84	665.01	2,587.95
Peak CUDA memory per rank (GB)	31.60	31.60	38.44

Table 3: Same global document schedule and 128 updates. Loop timing includes checkpoint I/O and final hashing, but excludes seed verification, model loading and distributed initialization. Complete process resource logs and instance lifetimes are retained separately.

Two g6e.xlarge hosts, each with one L40S, four vCPUs and 32 GiB RAM, communicate within one availability zone over private TCP/NCCL. The pair's updates take 4.39 times as long as the clean single-GPU baseline. Loop wall time is 1.31 times as long and allocated GPU loop time is 2.62 times as large. The single-GPU controls also show substantial checkpoint-I/O variation; these are single-run observations rather than stable hardware benchmarks. The two interfaces transmit 890.33 GB and 884.94 GB during the measured loop, about 1.775 TB combined sent traffic. Whole-interface counters include management traffic; received bytes are not counted again as independent transfers. Synchronizing roughly 6.85 GB of FP32 gradients each update creates a substantial communication cost.

Both ranks finish with exactly the same parameter digest, recorded in the public candidate receipt. One-rank and two-rank parameters differ because reduction order changes floating-point rounding: relative L2 difference is 0.000322 and maximum absolute difference 0.000801 across 218 tensors. Task quality and within-group byte agreement are separate measurements. This training uses complete model/optimizer replicas on each GPU, so it does not demonstrate memory pooling, model growth, independent ownership or cheap verification against a malicious worker. No full-model training failure recovery is tested in this run.

Serving uses the shared checkpoint on both hosts. After separate four-request warmups, the fixed workload runs once per phase, in order. Two replicas provide 1.67 times aggregate throughput. Stopping the second serving process before the failure workload causes 16 first-attempt connection

Serving phase	Completed	Wall time (s)	Requests/s	p95 latency (s)
One replica	32/32	11.02	2.90	0.873
Two replicas	32/32	6.61	4.84	0.663
One of two stopped	32/32	11.01	2.91	0.883

Table 4: Same shared checkpoint, fixed development tasks, client concurrency two and fresh request identities. Completed requests are not a correctness score. All corresponding output token sequences agree, with zero cache hits.

failures and 16 successful retries on the survivor, retaining original request identities. This is an immediate service-unavailability test, not a slow partition or a failure between execution and acknowledgement. Private fixed task IDs, one client and a volatile retry cache do not establish public provider discovery, long-conversation capacity or durable exactly-once billing.

The experiment has a $100 ceiling, an eight-hour automatic stop deadline and a compute rate of $1.861 per host-hour. All final model weights are copied and hash-verified; the full clean-single optimizer/RNG checkpoint is retained, while public model archives omit optimizer state. Both GPU instances and their root volumes are removed after evidence preservation; the public resource receipt records lifetime-based compute estimates separately from storage, transfer, IPv4 and tax. Root-device read-ahead is 128 KiB during all training and changes to 4,096 KiB afterward for staging and evaluation, without changing numerical source.

The result motivates testing less frequent training communication with explicit aggregation semantics and a fresh quality comparison. It also motivates stronger arithmetic targets and broader retention tests before assistant-facing promotion. This run issues no NEURO, performs no native GPU training or promotion transaction, and does not change consensus. Complete evidence and downloadable weights are at https://neuroshard.com/experiments/cooperative-learning-20260913/.

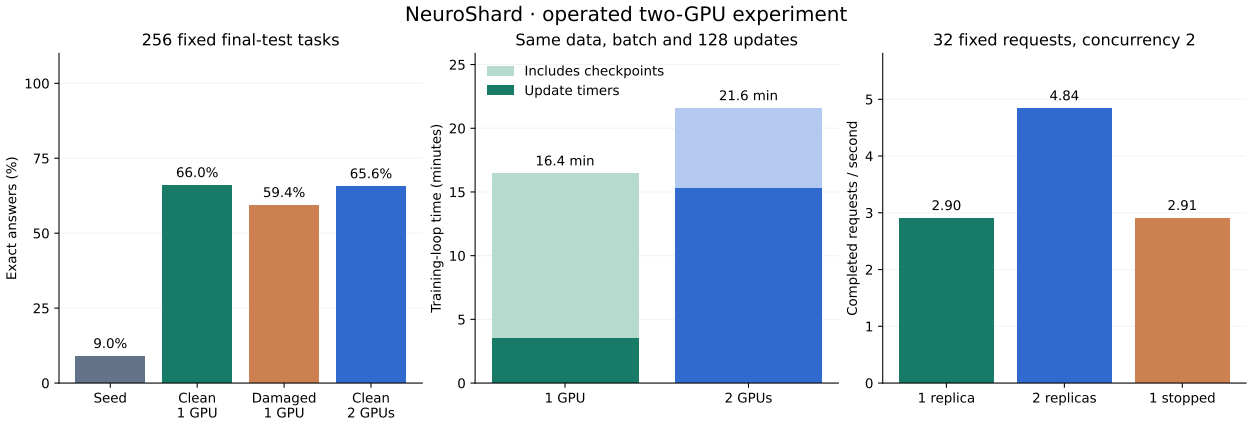


Figure 3: Measured task accuracy, training-loop cost and serving throughput. Task learning and service replication succeed within the stated scope; synchronous training has substantial overhead on these two hosts. Training-loop timing excludes initial loading and distributed setup.

16 Four-worker local training and coordinated recovery

The next operated experiment addresses the communication cost of the previous two-GPU run. Three predetermined arms start from the same pinned SmolLM2-1.7B-Instruct seed and train all

1,711,376,384 parameters: single-GPU AdamW, four-worker synchronous DDP, and four-worker local AdamW with an outer Nesterov update. Each processes the same 1,024-document schedule in 128 updates, at global batch eight. A worker processes two documents per update in either four-worker arm. Parameters, gradients and inner Adam states remain FP32; CUDA forward/backward uses BF16 autocast.

The local method follows the structure of DiLoCo [2]. For common parent x, local endpoint y_r, $W = 4$, outer rate $\eta = 0.7$ and momentum $\mu = 0.9$, each window applies

$$d = \frac{1}{W} \sum_{r=0}^{W-1} (x - y_r), \qquad v \leftarrow \mu v + d, \qquad x \leftarrow x - \eta(d + \mu v).$$

Workers replace their parameters with the new x every 16 local updates, while retaining their own Adam moments. The outer parent and velocity live on CPU, and FP32 deltas are reduced in chunks no larger than 32 MiB. Each local loss uses its worker's weighted target count. This differs from DDP's global token normalization when response lengths differ. Algorithmic trajectory and numerical reduction order therefore change; quality preservation is measured rather than assumed.

The data contain 768 fresh generated tasks and the same 256 published conversation-replay records. Grounded response tokens have weight eight; replay tokens have weight one. New task identities and prompts exclude the previous experiment's train, development and test cases. The final test has 1,024 cases across the same four families, including wording variants absent from training. Development has 64 separate cases. Retention uses the same 128 previously exposed public response-loss probes, so it cannot certify broad retention. Input and numerical-source receipts are committed before training, and all three predetermined final candidates are committed before final-test generation. No candidate is selected by final-test score.

The primary local-window candidate must gain at least five percentage points against the seed, with an exact one-sided paired $p \leq 0.01$, and its approximate paired 99% upper retention loss-change bound must be at most $+0.02$ nats. Its approximate paired 99% lower accuracy-change bound against four-worker DDP must be at least -0.03. These are single-run, narrow-distribution criteria. The efficiency criteria separately require active training time at most one half and transmitted bytes at most one eighth of DDP. Active time includes updates and synchronization; checkpoint, startup and whole-instance allocation costs are reported separately.

Each rank writes a durable model, local optimizer/RNG state, assignment journal and outer velocity before the group commits a manifest containing every rank's receipt. Restart chooses the latest identical manifest actually present on all workers and verifies each local checkpoint. A local latest pointer alone cannot advance the group. Rank belongs to the checkpoint identity, preventing substitution of another worker's Adam moments. Uncommitted outputs beyond the common point are retained as orphaned attempts. An OS lock rejects overlapping GPU jobs from the same Unix user before allocation.

The recovery trial restores copies of the uninterrupted step-64 group checkpoint, retaining rank identities and original deadlines. A supervisor kills rank one with SIGKILL after observing completed step 75. The group must fail within a bounded interval, then restart from step 64 with the same membership. Host-side comparisons check exact final model tensors, outer velocity, every rank's Adam/RNG tensors and scalars, and the numerical trajectory. Serialized file hashes need not be canonical. This is coordinated recovery under one operator, not elastic or Byzantine recovery.

Four g6e.xlarge hosts each provide one L40S, four vCPUs and 32 GiB host RAM. Capacity requires three hosts in us-east-1c and one in us-east-1d. All use 300 GiB encrypted gp3 disks, provisioned at 500 MiB/s and 6,000 IOPS, with 4,096 KiB read-ahead. The instance EBS baseline is 125 MB/s despite the higher provisioned volume throughput. AWS reports a 2.5 Gbit/s baseline

and up-to-20-Gbit/s burst bandwidth for this size. Network allowance counters and per-update timings are retained. The arms run in fixed order; network credits can be consumed during DDP and replenished during checkpoint pauses. Observed wall-time ratios are consequently not isolated causal estimates of algorithmic speedup. Whole-interface transmitted bytes, actual process durations and allocation costs remain separate measurements.

An initial launcher incorrectly treated the return of a detached session as job completion, overlapping arms and exhausting GPU memory. The attempts were stopped and preserved before candidate selection or final scoring. The corrected launcher waits for the child and propagates its exit status; a live competing process test verifies the new device lock. The inputs are recommitted against this corrected runner without changing the data, hyperparameters or quality criteria.

Serving uses the predetermined local-window final model independently of its test score. After identical development warmups, one replica, four replicas and four advertised endpoints with one stopped serve the same 64 development requests at concurrency four. Fresh request identities prevent warmup or retry-cache hits from inflating throughput. The outage trial is immediate process unavailability, not a slow partition or a crash between execution and acknowledgement. The fixture accepts fixed task IDs; it does not implement general private chat or native billing.

16.1 Measured learning, communication and recovery

The primary quality contract fails. Out of 1,024 test cases, the seed, single-GPU, four-worker DDP and local-window models answer 105, 641, 613 and 557 correctly, respectively. Local windows gain 454 answers and lose two against the seed, a +44.14-percentage-point change with exact one-sided paired $p = 5.60 \times 10^{-133}$. Against DDP they gain 75 and lose 131, a −5.47-point change; the approximate paired 99% lower bound is −9.05 points, outside the −3-point margin. Mean exposed retention loss changes by +0.083294 nats and its approximate 99% upper bound is +0.108140, exceeding +0.02. The corresponding upper bounds for the single-GPU and DDP controls are +0.006411 and +0.005859.

Local windows answer all 256 lookup cases correctly, but only 152 filtering and 149 sorting cases. Single-GPU training answers 247, 206 and 188; DDP answers 224, 223 and 166. Every model fails all 256 invoice totals. The perfect lookup result therefore conceals important specialization and retention tradeoffs. One generated-task run and exposed conversation probes cannot establish general assistant quality or continual learning.

Measurement	Single	Four DDP	Four local
Maximum active seconds	208.09	1,979.26	250.08
Maximum loop seconds	266.12	2,048.37	494.51
Complete process seconds	290.76	2,078.61	532.54
Allocated loop GPU seconds	266.12	8,193.20	1,978.02
Transmitted GB	0.000020	5,330.663	331.778

Local windows use 12.64% of DDP's active time and 6.22% of its transmitted bytes, passing the efficiency criteria. They remain 1.20 times slower in active time and 1.86 times slower through checkpoints than one GPU, with four GPUs allocated. DDP's median update rises from 7.76 seconds at steps 2–90 to 33.15 seconds at steps 91–128 while network allowance counters increase. These instance limits and fixed arm order qualify the observed timing ratio; communication reduction does not demonstrate an economical training advantage over one GPU or acceptable learning quality.

The killed rank is detected about 2.47 seconds after fault injection; surviving processes finish shutting down about 32.74 seconds after detection. The restarted group takes 357.04 seconds

including restore, repeated work and its final checkpoint. Every rank's final model, outer velocity, Adam/RNG tensors and scalars, and numerical trajectory exactly match the uninterrupted run. Large checkpoint file hashes also match. Full host-side comparisons take 199.80–220.41 seconds and overlap backup I/O; this cost is separate from restart time. This is process-crash recovery on the same hosts, not recovery from instance or disk loss, elastic admission or Byzantine behavior.

16.2 A separate compressed-gradient feasibility probe

After preserving the quality failure, a training-only probe evaluates rank-32 PowerSGD [1] with a common reconstructed gradient and replicated Adam states. Eight dense warmup updates precede four compressed updates. Dense FP32 error feedback and projection warm starts are enabled. The first full-model attempt completes step nine but exhausts CUDA memory on step ten. Its failed logs and source identity are retained.

A wrapper restores one bucket's dense error from CPU immediately before the pinned upstream hook, completes its collective sequence, then copies the new residual back to CPU. Projection state remains on GPU. This adds PCIe transfers and sacrifices overlap; no residual quantization or new arithmetic is introduced. Small-model tests on CPU and on both real CUDA hosts compare model, Adam, error feedback, projections, loss and RNG states exactly against a serialized resident control, and against the unmodified upstream hook for a single bucket. The full 1.7B probe then completes all 12 updates on both GPUs with a common parameter digest and maximum CUDA allocation of 36.89 GiB.

The hook's counters report 45.14-fold logical compression during the four compressed updates, excluding dense warmup, initial broadcast and transport overhead. Full process time is 106.49 seconds with operator artifact I/O running concurrently. This is memory and transport feasibility, not a matched sustained-speed or quality result. No new holdout is scored, no probe checkpoint is selected, and no recovery codec for compression state is implemented. A subsequent full-length comparison requires fresh committed test cases, broader retention and complete compression-state recovery.

All three predetermined main candidates and all final answers remain in the public evidence. The operated GPU experiment does not issue NEURO, change consensus or promote a native serving model. Failed quality gates remain failed.

16.3 Serving and complete artifact preservation

The one-provider, four-provider and outage phases complete all 64 requests each, at 3.008, 9.025 and 6.713 requests per second. Four providers deliver 3.00-fold aggregate throughput at the same client concurrency. The outage phase needs 16 retries. Output tokens agree across all phases and no response is a cache hit. This measures service capacity for the predetermined failed-quality fixture; it does not approve that model for native serving.

All three model archives are publicly retained with member hashes. The complete original and recovered state inventory has 267 references and 177.992 GB of unique objects in hash-verified operator S3 backups. After 95.843 GB passed through the coordinator, remaining artifacts upload directly from workers using short-lived object/part capabilities and full readback. Secret AWS keys remain on the coordinator. These trusted operator storage receipts are not permissionless availability proofs. Bulk transfers finish before serving warmups and timings.

The four temporary GPU instances, root volumes and dedicated security group are removed after preservation. Compute through confirmed cleanup is at most $16.97; conservative transfer accounting and a $10 storage/setup allowance give $88.69 before credits, within the $100 allocation

plan. This is not a final AWS invoice and excludes later retained-storage and public-download charges. The existing native CPU network continues independently.

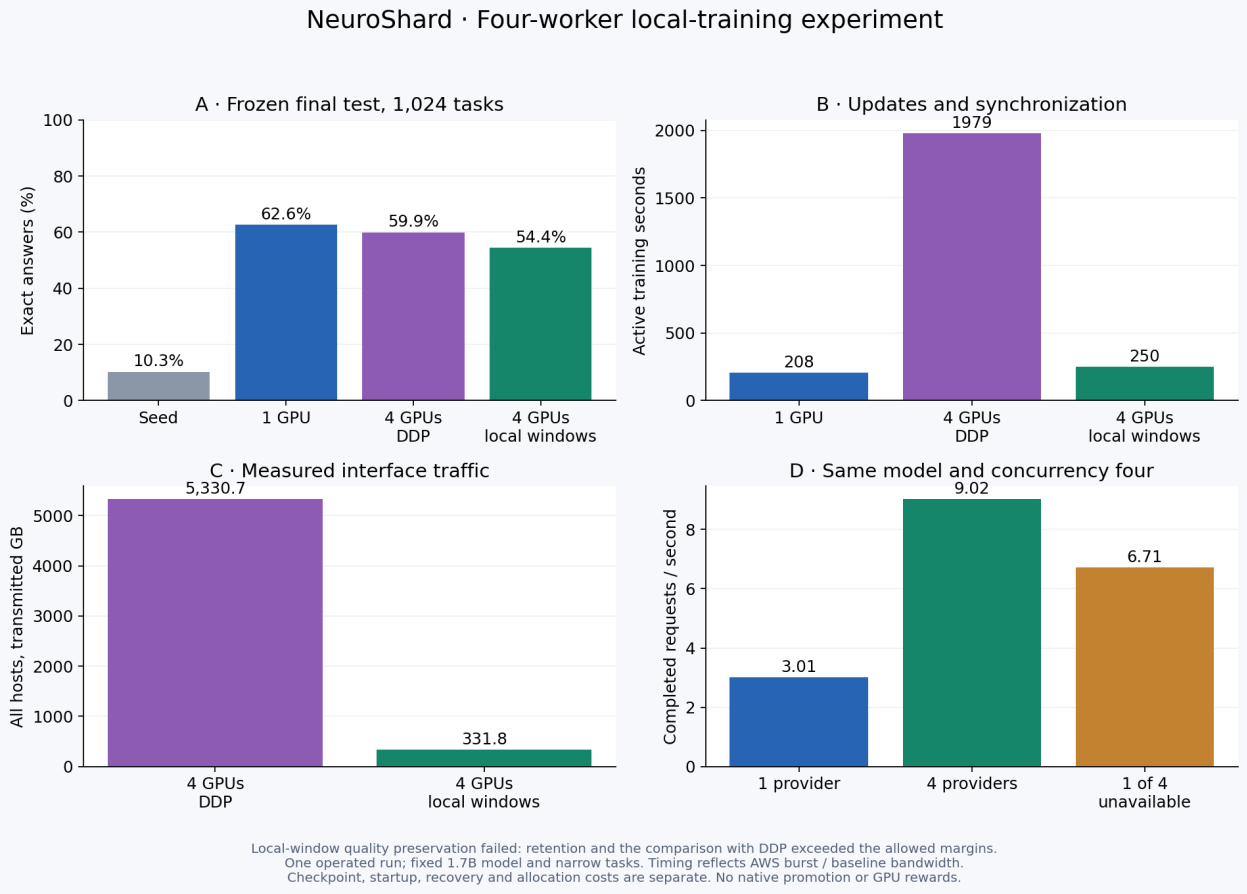


Figure 4: Four-worker measurements. Local windows reduce communication and recover exactly, but fail retention and the quality margin against DDP. Inference uses the same predetermined model and client concurrency in every phase. Timing depends on the operated AWS bandwidth regime; complete allocation and checkpoint costs are reported separately.

17 Scaling by useful capacity

17.1 Compute groups and growth admission

The architectural direction is a native ledger coordinating compute groups that jointly hold and execute a model. First allocate new resources to missing replicas, complete audit coverage, serving and training throughput. Additional parameters are justified only by sustained usable resources and measured benefit. Declared peer counts and memory limits are not proofs of capacity, reliability or independent ownership.

The first multi-group learning profile should use bounded synchronous windows from a common parent, with data, optimizer, seeds, contribution weights and deadlines fixed before execution. Its outer aggregation and reduction order need their own reproducible disputes. Missing contributors follow a precommitted membership or cancellation rule; a coordinator cannot invent their updates.

Asynchronous stale updates and sparse experts remain subsequent research profiles. Decoupled DiLoCo supplies useful evidence about failure isolation and wide-area cooperative training [11], but does not resolve public Byzantine admission or verification. Its public goodput comparison is simulated; the multi-region training run is separate experimental evidence.

Growth admission must budget training, complete replay, evaluation, artifact retention and serving after promotion. It needs a tested recovery path for the loss of a major failure domain, sustained service measurements and an equal-resource comparison with continued training of the smaller model. Growth creates no emission by itself. The former serving model remains available if added capacity disappears or the candidate is rejected.

17.2 Honest auditing as an explicit liability

For a work window, reserve

$$B = B_{\text{train}} + B_{\text{audit}} + B_{\text{availability}} + B_{\text{evaluation}} + B_{\text{settlement}}.$$

Any issuance component is part of a fixed period cap; sponsorship and inference payments transfer existing tokens. These are separate liabilities, with unspent funds returned to their sources. An audit allocation must remain payable for a completed correct service when the producer is honest. Repeated independent replay can be purchased from that allocation without creating another training-work emission.

This accounting is necessary but insufficient. A signature, matching root, commitment before public disclosure or key-specific hash cannot establish that independent machines performed separate replays: producers and auditors can share results privately. A first purchased complete-audit service must therefore expose operator and coverage assumptions rather than claim to solve the verifier's dilemma. A fraud-only observer with positive replay cost C, successful rewarded detection probability p, and bounty R receives expected net reward $pR - C$, which fails as p approaches zero for fixed R. Rewarding an unaudited "success" signature does not repair it.

Under independent auditor selection with adversarial resource fraction a, the probability that all k selected auditors are adversarial is a^k. Under fully correlated ownership it can remain a. For $a = 1/4$ and $k = 4$, these values are approximately 0.39% and 25%. They are illustrative conditional models, not measured failure probabilities of the current single-operator network. Random selection itself needs an analyzed beacon: a proposer-influenced block hash is not automatically an unbiased lottery.

These are explicit design obligations, not new claims about the deployed reward split. Before public activation, require funded complete coverage, objective false-attestation disputes, collateral and escrow conservation, analysis of copying and collusion, and independently operated services.

18 Implementation boundary and public-release requirements

The executable components include full-backbone pipeline training, durable worker operations, identity depth growth, fresh/replay data windows, response-target masks, paired evaluation, a local continual-epoch controller, bounded-stage replay, native reservation/reward settlement, collateralized availability/fraud challenges, and distributed generation. Their scopes and tests are recorded with the source and experiment artifacts.

The default experimental application retains its original pinned-batch behavior. Its opt-in lifecycle profile adds bonded rolling-data admission, finite fresh/replay training, verifiable multi-window metrics, native serving decisions, and paid generation. The funded extension adds prepaid

replay, audit collateral and reporting windows, a durable signing outbox, a recovering continuous operator and full-node onboarding. Worker RPC still uses operator-managed authentication over SSH. Public discovery, independent worker collateral, sustainable audit payments, durable model replication and garbage collection, efficient long-lived ledger state, and an explicit 0.4.0 migration need implementation and testing before public cutover. The current source commitment covers all repository Python modules and validators check numerical-library versions at startup. Further runtime conformance must cover platform and dependency behavior beyond version labels.

A responsible release gate is therefore concrete: a new participant joins through the public client; contributes prescribed full-model work; survives a replacement worker or coordinator; a corrupt update and withheld artifact fail under independent observers; new data and a larger eligible model activate through native transitions; an old balance buys inference from the promoted checkpoint; and migration preserves history without duplicate token claims. None of these missing integration steps is made true by calling an off-chain research registry decentralized.

19 What the protocol can promise

NeuroShard can specify an ongoing procedure for proposing, executing, checking, comparing, and serving model revisions. It can preserve a usable model while rejecting unsuccessful experiments. It can add capacity when resources and measured benefit justify it. Under explicit consensus, observer, availability, and evaluation assumptions, it can settle those actions and their payments.

It cannot promise that every batch, every additional parameter, or every calendar day makes an LM universally better. Data can become exhausted, a task can saturate, participants can leave, and metrics can be gamed. The defensible objective is continual verified experimentation with demonstrated improvements and controlled promotion. An integrated native epoch is a necessary engineering milestone. Independent public participation, economical verification, sustained learning evidence, and operational recovery must follow; none constitutes autonomous general intelligence by definition.

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